Canvas Quiz 3, Modules 4–5 (graded)

Module 4 - Drug Elimination and Clearance · Canvas quiz · 23 Sep 2026

Name _____ Date _____

Question 1 Which term describes the process by which drug is chemically converted in the body?

A. biotransformation
B. distribution
C. excretion
D. absorption

Question 2 If there is an increase in the total body clearance of an agent due to renal dysfunction, what is the expected change to the elimination half-life of that agent?

A. The elimination half-life will increase.
B. The elimination half-life will decrease.
C. The elimination half-life will not change.

Question 3 Which statement is true regarding the time needed to reach the maximum concentration following administration of an oral dose?

A. the time needed to reach the maximum concentration depends on the rate constants for absorption and elimination
B. the time needed to reach the maximum concentration increases if the dose is decreased
C. the time needed to reach the maximum concentration increases if the dose is increased
D. the time needed to reach the maximum concentration is independent of the rate constants for absorption and elimination

Question 4 Which of the following is true regarding the C_{max} following oral administration?

A. the rate of drug absorption equals the rate of drug elimination
B. the rate of drug elimination is faster than the rate of drug absorption
C. drug at the absorption site has become depleted
D. the rate of drug absorption is faster than the rate of drug elimination

Question 5 When does the maximum concentration of drug in the plasma occur following a single 1000-mg oral dose of an agent that has an oral bioavailability of 83%, apparent volume of distribution of 18 L, absorption half-life of 0.8 hours, and elimination half-life of 5.6 hours? Round to the nearest tenth (one decimal place).

Question 6 A 500-mg IV bolus dose of an antibiotic with an apparent volume of distribution of 23 L and half-life of 6.0 hr was administered to a 75 kg male volunteer. What is the expected AUC of this dose? Round to the nearest tenth (one decimal place).

Question 7 1,000 mg of a drug was given by IV bolus injection. The V_D is 19 L and the elimination half-life is 3 hours. Urine samples were collected for 48 hours and 713 mg of unchanged drug was recovered. What is the renal clearance of this drug? Round to the nearest hundredth (two decimal places).

Question 8 The kinetics following the oral administration of a single 500-mg dose of a medicinal agent best fits a one-compartment model described by the equation $C_p = 30(e^{-0.133t} - e^{-0.934t})$. What is the volume of distribution following the 500-mg dose if the oral bioavailability is 82%? Assume units of mcg/mL for C_p and hr for time. Round to the nearest tenth (one decimal place).